

Strength of Materials II

Chapter Four: Energy Method

MEng3111



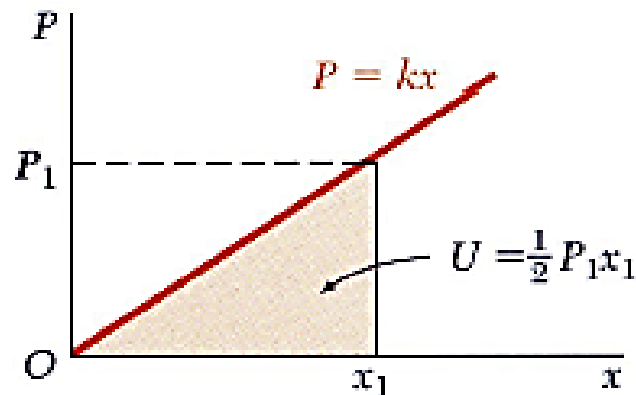
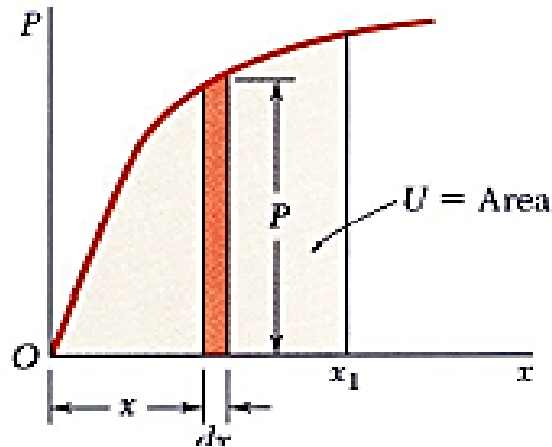
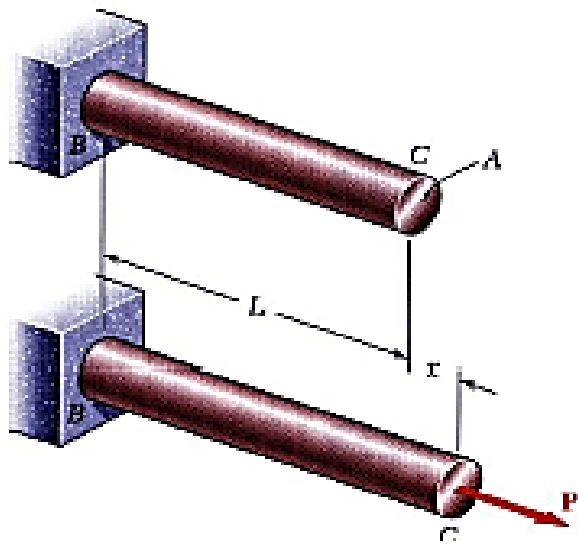
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Energy Method

Learning Objectives

- **Compute** the strain energy due to axial, bending, and torsion loading.
- **Define** the work done by a force or couple.
- **Determine** displacements from a single load using the workenergy method.
- **Apply** Castigliano's theorem to determine displacements due to multiple loads.
- **Solve** statically indeterminate problems using Castigliano's theorem.

Energy Method



- A uniform rod is subjected to a slowly increasing load
- The *elementary work done* by the load P as the rod elongates by a small dx is

$$dU = P dx = \text{elementary work}$$

- which is equal to the area of width dx under the load-deformation diagram.
- The *total workdone* by the load for a deformation x_1

$$\text{Strain energy} = U = \int_0^{x_1} P dx \quad \text{N}\cdot\text{m or joule (J)}$$

- which results in an increase of *strain energy* in the rod.
- In the case of a **linear elastic deformation**

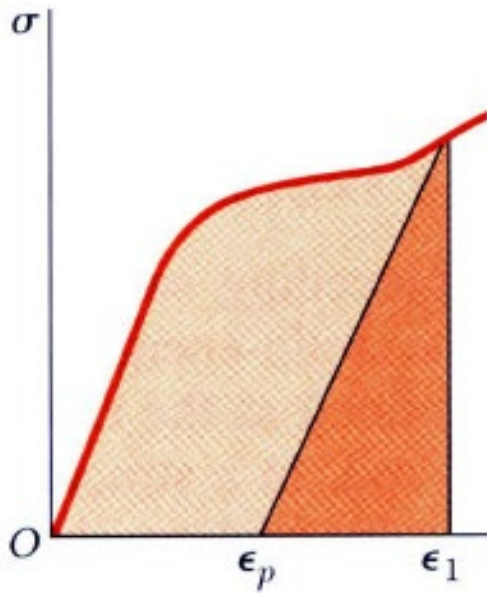
$$U = \int_0^{x_1} kx dx = \frac{1}{2} kx_1^2 = \frac{1}{2} P_1 x_1$$

Strain Energy Density

- To eliminate the effects of size, evaluate the strain-energy per unit volume,

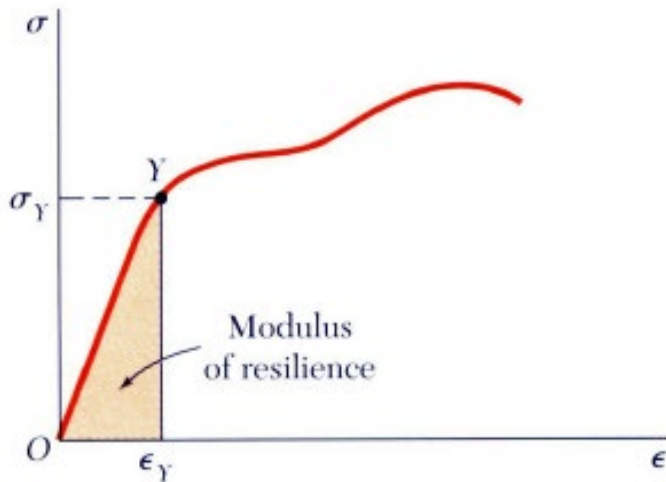
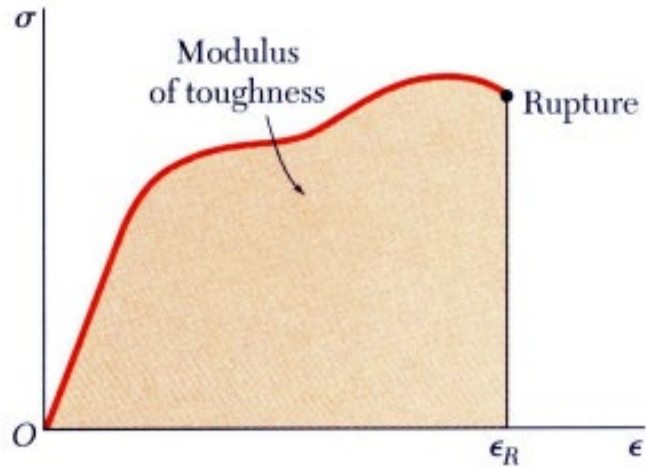
$$\frac{U}{V} = \int_0^{x_1} \frac{P}{A} \frac{dx}{L}$$

$$u = \int_0^{\epsilon_1} \sigma_x d\epsilon = \text{strain energy density}$$



- The total strain energy density resulting from the deformation is equal to the area under the curve to ϵ_1 .
- As the material is unloaded, the stress returns to zero but there is a permanent deformation. Only the strain energy represented by the triangular area is recovered.
- Remainder of the energy spent in deforming the material is dissipated as heat.

Strain Energy Density



- The strain energy density resulting from setting $\epsilon_1 = \epsilon_R$ the **modulus of toughness**.
- The energy per unit volume required to cause the material to rupture is related to its ductility as well as its ultimate strength.
- If the stress remains within the proportional limit,

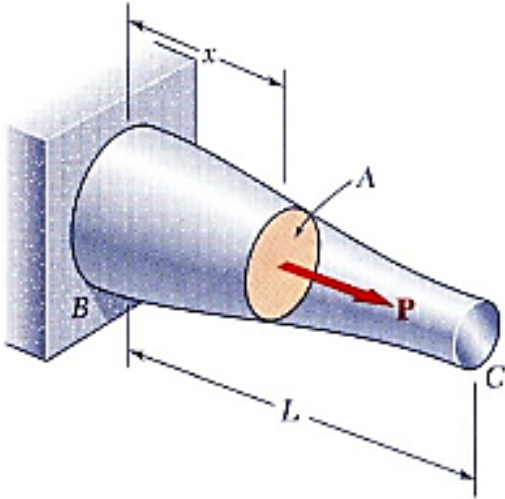
$$u = \int_0^{\epsilon_1} E \epsilon_x d\epsilon_x = \frac{E \epsilon_1^2}{2} = \frac{\sigma_1^2}{2E}$$

- The strain energy density resulting from setting $\sigma_1 = \sigma_Y$ is the **modulus of resilience**.

$$u_Y = \frac{\sigma_Y^2}{2E} = \text{modulus of resilience}$$

Elastic Strain Energy for Normal Stresses

Axial Loading



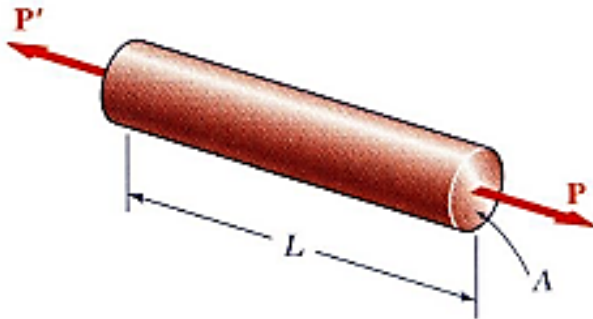
- In an element with a nonuniform stress distribution

$$u = \lim_{\Delta V \rightarrow 0} \frac{\Delta U}{\Delta V} = \frac{dU}{dV} \quad U = \int u dV = \text{total strain energy}$$

- For values of $u < u_y$, i.e., below the proportional limit,

$$U = \int \frac{\sigma_x^2}{2E} dV = \text{elastic strain energy}$$

- Under axial loading, $\sigma_x = P/A$ $dV = A dx$

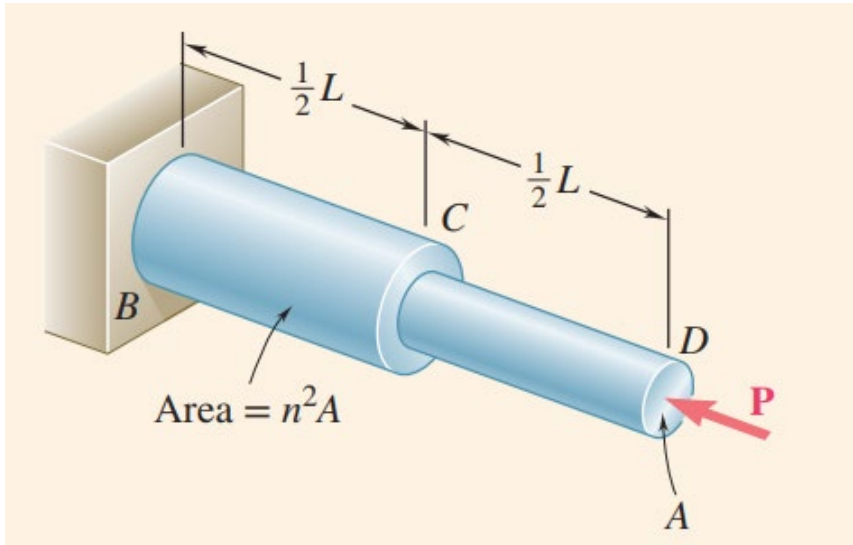


$$U = \int_0^L \frac{P^2}{2AE} dx$$

- For a rod of uniform cross-section,

$$U = \frac{P^2 L}{2AE}$$

Concept Application



$$U = \frac{P^2 L}{2AE}$$

A rod consists of two portions BC and CD of the same material, same length, but of different cross sections. Determine the strain energy of the rod when it is subjected to a centric axial load $\mathbf{P}$, expressing the result in terms of P , L , E , the cross-sectional area A of portion CD , and the ratio n of the two diameters.

$$U_n = \frac{P^2(\frac{1}{2}L)}{2AE} + \frac{P^2(\frac{1}{2}L)}{2(n^2A)E} = \frac{P^2 L}{4AE} \left(1 + \frac{1}{n^2} \right)$$

Or

$$U_n = \frac{1 + n^2}{2n^2} \frac{P^2 L}{2AE}$$

Check that, for $n = 1$

$$U_1 = \frac{P^2 L}{2AE}$$

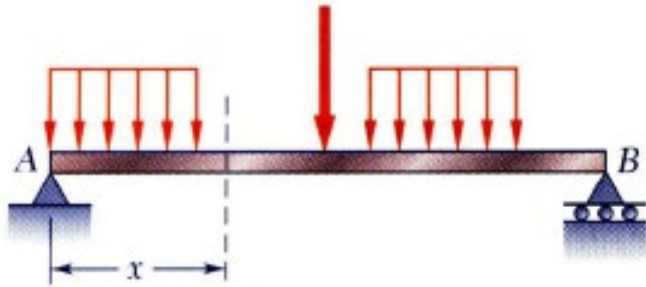
Also note that for $n > 1$, $U_n < U_1$.

As an example, when

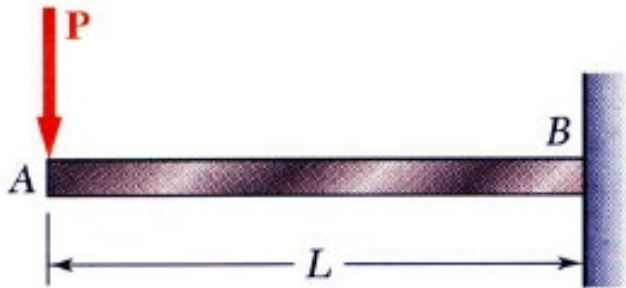
$n = 2$, $U_2 = (5/8)U_1$. Since the maximum stress occurs in portion CD of the rod and is equal to $\sigma_{\max} = P/A$, then for a given allowable stress, increasing the diameter of portion BC of the rod results in a *decrease* of the overall energy-absorbing capacity. Unnecessary changes in cross-sectional area should be avoided in the design of members subjected to loads (such as impact loadings) where the energy-absorbing capacity of the member is critical.

Elastic Strain Energy for Normal Stresses

Bending loading



$$\sigma_x = \frac{My}{I}$$



- For a beam subjected to a bending load.

$$U = \int \frac{\sigma_x^2}{2E} dV = \int \frac{M^2 y^2}{2EI^2} dV$$

- Setting $dV = dx \int dA$

$$\begin{aligned} U &= \int_0^L \int_A \frac{M^2 y^2}{2EI^2} dA dx = \int_0^L \frac{M^2}{2EI^2} \left(\int_A y^2 dA \right) dx \\ &= \int_0^L \frac{M^2}{2EI} dx \end{aligned}$$

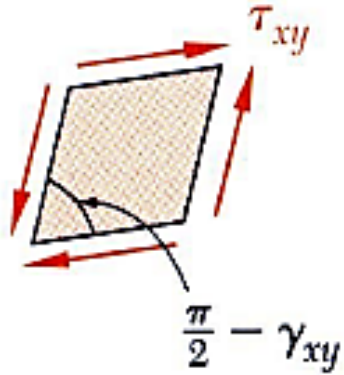
- For an end-loaded cantilever beam

$$M = -Px$$

$$U = \int_0^L \frac{P^2 x^2}{2EI} dx = \frac{P^2 L^3}{6EI}$$

Strain Energy For Shearing Stresses

- For a material subjected to plane shearing stresses



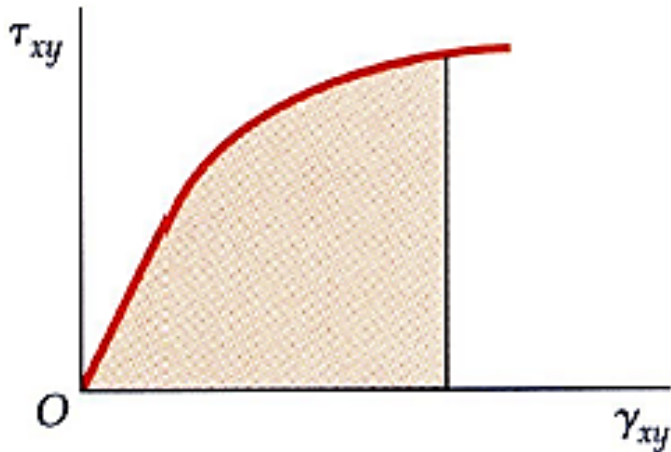
$$u = \int_0^{\gamma_{xy}} \tau_{xy} d\gamma_{xy}$$

- For value of γ_{xy} within the proportional limit

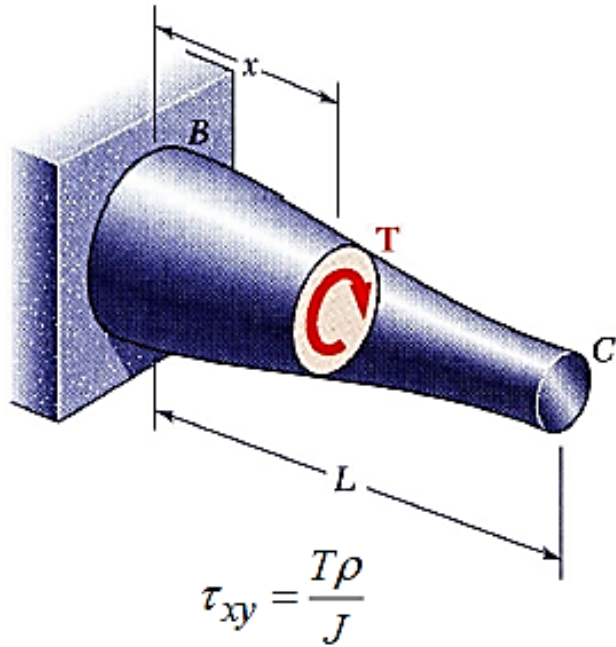
$$u = \frac{1}{2} G \gamma_{xy}^2 = \frac{1}{2} \tau_{xy} \gamma_{xy} = \frac{\tau_{xy}^2}{2G}$$

- The total strain energy is found from

$$\begin{aligned} U &= \int u dV \\ &= \int \frac{\tau_{xy}^2}{2G} dV \end{aligned}$$



Strain Energy For Shearing Stresses

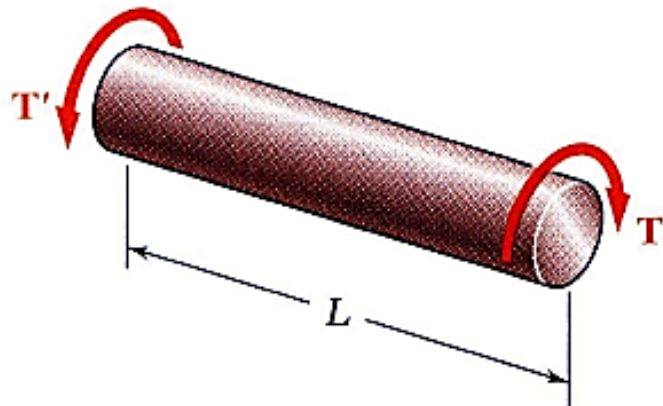


$$U = \int \frac{\tau_{xy}^2}{2G} dV = \int \frac{T^2 \rho^2}{2GJ^2} dV$$

- Setting $dV = dx \int dA$,

$$U = \int_0^L \int_A \frac{T^2 \rho^2}{2GJ^2} dA dx = \int_0^L \frac{T^2}{2GJ^2} \left(\int_A \rho^2 dA \right) dx$$

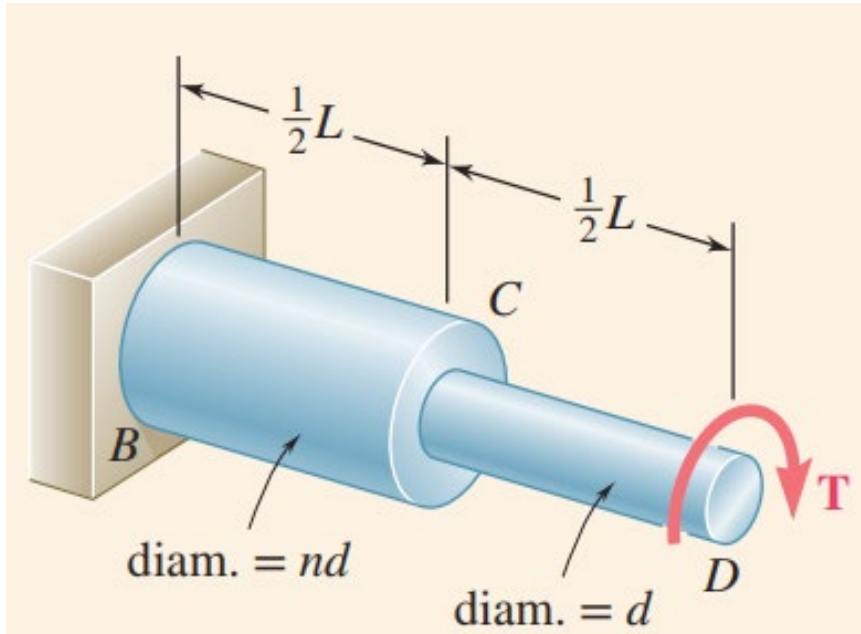
$$= \int_0^L \frac{T^2}{2GJ} dx$$



- In the case of uniform

$$U = \frac{T^2 L}{2GJ}$$

Concept Application



A circular shaft consists of two portions BC and CD of the same material and

length, but of different cross sections

Determine the strain energy of the shaft when it is subjected to a twisting couple T at end D . Express the results in terms of T , L , G , the polar moment of inertia J of the smaller cross section, and the ratio n of the two diameters.

$$U_n = \frac{T^2(\frac{1}{2}L)}{2GJ} + \frac{T^2(\frac{1}{2}L)}{2G(n^4J)} = \frac{T^2L}{4GJ} \left(1 + \frac{1}{n^4} \right)$$

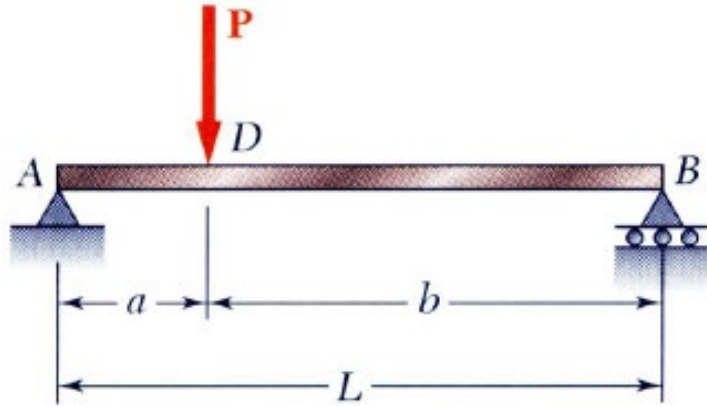
or

$$U_n = \frac{1 + n^4}{2n^4} \frac{T^2L}{2GJ}$$

For $n = 1$,

$$U_1 = \frac{T^2L}{2GJ}$$

Example:



- a) Taking into account only the normal stresses due to bending, determine the strain energy of the beam for the loading shown.
- b) Evaluate the strain energy knowing that the beam is a W10x45, $P = 40$ kips, $L = 12$ ft, $a = 3$ ft, $b = 9$ ft, and $E = 29 \times 10^6$ psi.

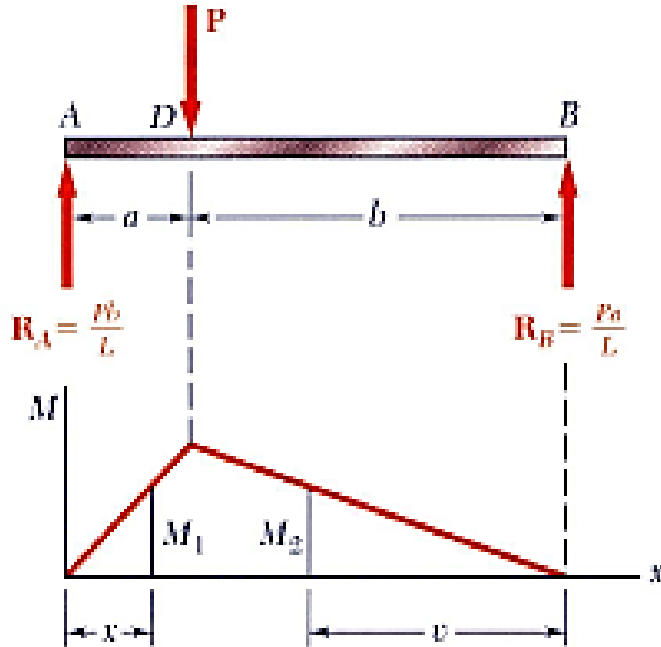
SOLUTION:

- Determine the reactions at A and B from a free-body diagram of the complete beam.
- Develop a diagram of the bending moment distribution.
- Integrate over the volume of the beam to find the strain energy.
- Apply the particular given conditions to evaluate the strain energy.

Example:

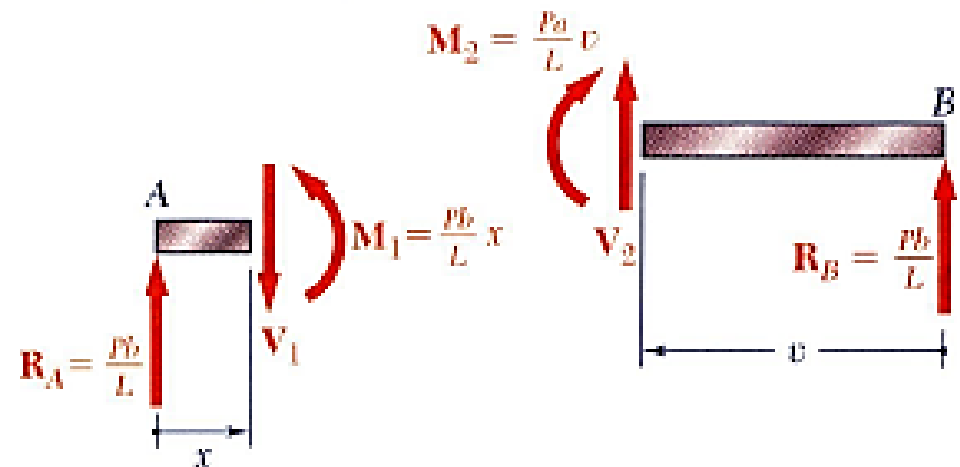
SOLUTION:

Determine the reactions at A and B from a free-body diagram of the complete beam.



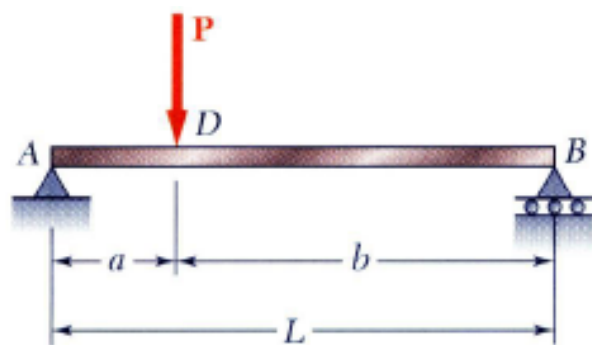
$$R_A = \frac{Pb}{L} \quad R_B = \frac{Pa}{L}$$

Develop a diagram of the bending moment distribution.



$$M_1 = \frac{Pb}{L}x \quad M_2 = \frac{Pa}{L}v$$

Example:



Over the portion AD,

$$M_1 = \frac{Pb}{L}x$$

Over the portion BD,

$$M_2 = \frac{Pa}{L}v$$

$$P = 45 \text{ kips} \quad L = 144 \text{ in.}$$

$$a = 36 \text{ in.} \quad b = 108 \text{ in.}$$

$$E = 29 \times 10^3 \text{ ksi} \quad I = 248 \text{ in}^4$$

- Integrate over the volume of the beam to find the strain energy.

$$\begin{aligned} U &= \int_0^a \frac{M_1^2}{2EI} dx + \int_0^b \frac{M_2^2}{2EI} dv \\ &= \frac{1}{2EI} \int_0^a \left(\frac{Pb}{L}x \right)^2 dx + \frac{1}{2EI} \int_0^b \left(\frac{Pa}{L}x \right)^2 dx \\ &= \frac{1}{2EI} \frac{P^2}{L^2} \left(\frac{b^2 a^3}{3} + \frac{a^2 b^3}{3} \right) = \frac{P^2 a^2 b^2}{6EIL^2} (a + b) \end{aligned}$$

$$U = \frac{P^2 a^2 b^2}{6EIL}$$

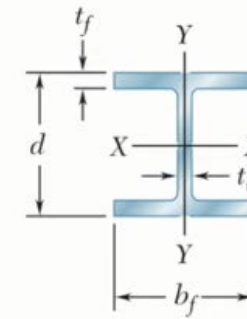
$$U = \frac{(40 \text{ kips})^2 (36 \text{ in})^2 (108 \text{ in})^2}{6(29 \times 10^3 \text{ ksi})(248 \text{ in}^4)(144 \text{ in})}$$

$$U = 3.89 \text{ in} \cdot \text{kips}$$

Appendix C Properties of Rolled-Steel Shapes

(U.S. Customary Units)
Continued from page A18

W Shapes (Wide-Flange Shapes)



Designation [†]			Flange		Web Thick- ness t_{wr} in.	Axis X-X			Axis Y-Y		
			Width b_f in.	Thick- ness t_f in.		$I_{x'} \text{ in}^4$	$S_{x'} \text{ in}^3$	$r_{x'} \text{ in.}$	$I_{y'} \text{ in}^4$	$S_{y'} \text{ in}^3$	$r_{y'} \text{ in.}$
W12 × 96	28.2	12.7	12.2	0.900	0.550	833	131	5.44	270	44.4	3.09
72	21.1	12.3	12.0	0.670	0.430	597	97.4	5.31	195	32.4	3.04
50	14.6	12.2	8.08	0.640	0.370	391	64.2	5.18	56.3	13.9	1.96
40	11.7	11.9	8.01	0.515	0.295	307	51.5	5.13	44.1	11.0	1.94
35	10.3	12.5	6.56	0.520	0.300	285	45.6	5.25	24.5	7.47	1.54
30	8.79	12.3	6.52	0.440	0.260	238	38.6	5.21	20.3	6.24	1.52
26	7.65	12.2	6.49	0.380	0.230	204	33.4	5.17	17.3	5.34	1.51
22	6.48	12.3	4.03	0.425	0.260	156	25.4	4.91	4.66	2.31	0.848
16	4.71	12.0	3.99	0.265	0.220	103	17.1	4.67	2.82	1.41	0.773
W10 × 112	32.9	11.4	10.4	1.25	0.755	716	126	4.66	236	45.3	2.68
68	20.0	10.4	10.1	0.770	0.470	394	75.7	4.44	134	26.4	2.59
54	15.8	10.1	10.0	0.615	0.370	303	60.0	4.37	103	20.6	2.56
45	13.3	10.1	8.02	0.620	0.350	248	49.1	4.32	53.4	13.3	2.01
39	11.5	9.92	7.99	0.530	0.315	209	42.1	4.27	45.0	11.3	1.98
33	9.71	9.73	7.96	0.435	0.290	171	35.0	4.19	36.6	9.20	1.94
30	8.84	10.5	5.81	0.510	0.300	170	32.4	4.38	16.7	5.75	1.37
22	6.49	10.2	5.75	0.360	0.240	118	23.2	4.27	11.4	3.97	1.33
19	5.62	10.2	4.02	0.395	0.250	96.3	18.8	4.14	4.29	2.14	0.874
15	4.41	10.0	4.00	0.270	0.230	68.9	13.8	3.95	2.89	1.45	0.810

Strain Energy for a General State of Stress

- Previously found strain energy due to uniaxial stress and plane shearing stress. For a general state of stress.

$$u = \frac{1}{2} (\sigma_x \varepsilon_x + \sigma_y \varepsilon_y + \sigma_z \varepsilon_z + \tau_{xy} \gamma_{xy} + \tau_{yz} \gamma_{yz} + \tau_{zx} \gamma_{zx})$$

- With respect to the principal axes for an elastic, isotropic body

$$u = \frac{1}{2E} [\sigma_a^2 + \sigma_b^2 + \sigma_c^2 - 2\nu(\sigma_a \sigma_b + \sigma_b \sigma_c + \sigma_c \sigma_a)]$$

$$= u_v + u_d$$

$$u_v = \frac{1-2\nu}{6E} (\sigma_a + \sigma_b + \sigma_c)^2 = \text{due to volume change}$$

$$u_d = \frac{1}{12G} [(\sigma_a - \sigma_b)^2 + (\sigma_b - \sigma_c)^2 + (\sigma_c - \sigma_a)^2] = \text{due to distortion}$$

- Basis for the maximum distortion energy failure criteria

$$u_d < (u_d)_Y = \frac{\sigma_Y^2}{6G} \text{ for a tensile test specimen}$$



Thank you

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